



Lead with Trust. Inspire with Vision.

CUSTOMER SUCCESS STORY

CASE STUDY

Integrated IP-Based PA System & FLP Telephone Communication

Plant-wide communication modernization at Dahej, Gujarat

CLIENT



KRYSTAL INGREDIENTS PRIVATE LIMITED

Connecting Every Corner of the Plant

A unified communication platform designed for safer, faster and more dependable announcements.

Krystal Ingredients Private Limited required an integrated communication system that could use its existing Matrix ETERNITY GENX12SAC PABX while extending reliable public-address coverage to operational, administrative and MPP plant areas.

27

IP SPEAKERS

3

PAGING CONSOLES

12

FLP TELEPHONES

18

OUTDOOR SPEAKERS

The project at a glance

- Integrated the IP PA environment with the existing enterprise PABX.
- Enabled authorized users to initiate announcements from telephones and paging microphones.
- Configured zone-wise and all-call announcement access codes.
- Created a scalable foundation for future zones, speakers and plant expansion.

One Plant. Multiple Areas. One Communication

01 **Fragmented communication**

Plant teams needed a consistent method to reach personnel across administration, security, utilities and process areas.

02 **Hazardous-area requirements**

MPP plant communication required flameproof telephone deployment suitable for the operating environment.

03 **Existing infrastructure**

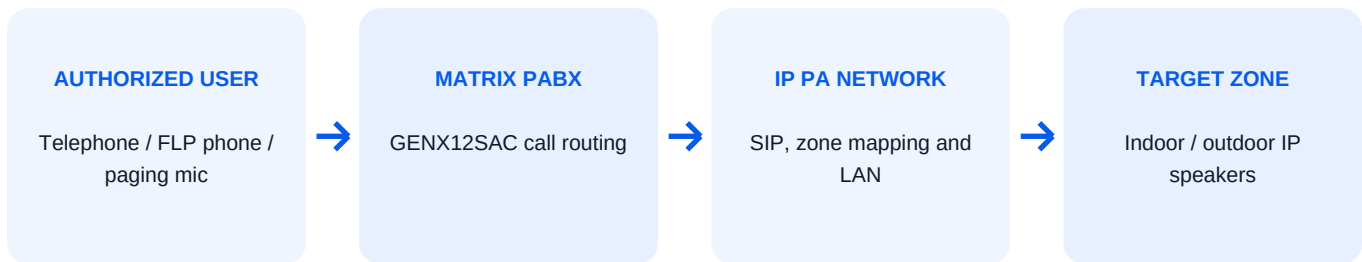
The solution had to work with the installed Matrix PABX and LAN, minimizing avoidable changes.

04 **Controlled announcements**

Management required authorized, zone-specific and plant-wide announcements using predefined codes.

From Telephone Call to Targeted Announcements

The design converts authorized calls into IP-based paging across predefined plant zones.



Configured communication paths

- Individual-zone calling for focused operational communication.
- All-call facility for urgent plant-wide announcements.
- PABX-based access control for authorized extensions.
- IP network foundation supporting centralized configuration and future expansion.

Coverage Across Critical Plant Areas

ADMIN & SECURITY

Admin GF/FF/SF, security, OHC and DCS

PROCESS & UTILITIES

Pesso Tank, warehouse, utility, boiler, ETP and MEE

MPP PLANT LEVELS

0 m, 6 m, 12 m, 19 m, 26 m and terrace

SPARES & RESILIENCE

Two indoor and one outdoor spare speaker

Installed system quantities

Indoor IP speakers	9
Outdoor IP speakers	18
Paging microphones	3
FLP telephones	12

A Structured Delivery Method

01

Survey & Engineering

Reviewed the PABX, IP network, telephone infrastructure and announcement requirements.

02

Installation

Deployed IP speakers, paging microphones, FLP telephones and associated connectivity.

03

Integration & Configuration

Programmed SIP connectivity, routing, access codes, user permissions and zone mappings.

04

Testing & Commissioning

Verified end-to-end calling, zone-wise paging, all-call operation and audio performance.

05

Documentation & Handover

Recorded addressing, device identification, configuration and operating guidance.

A Stronger Communication Backbone

The commissioned solution improves the way plant teams communicate across routine and urgent situations.



Faster reach

Authorized users can communicate from familiar telephones or paging consoles.



Targeted messaging

Zone mapping helps direct announcements only where they are needed.



Improved safety support

All-call capability supports rapid dissemination of critical information.



Lower complexity

Existing PABX and IP infrastructure are leveraged as part of the integrated design.



Scalable platform

New speakers, zones and plant areas can be incorporated as requirements grow.



Documented control

Device addressing, access codes and configuration parameters are recorded for support.



BUILDING SAFER, SMARTER PLANTS.

D-TECH delivers integrated enterprise technology and communication solutions engineered around real operational needs.

D-TECH SOLUTION INTEGRATORS PVT. LTD.

sales@dtechindia.com

+91 98980 64593 | www.dtechindia.com